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CORRESPONDENCE

Hypertensive heart disease: a structural and functional continuum rather than a coding construct

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To the Editor:

Hypertension Research

The editorial by Maria Giulia Bellicini [1] presents a provocative argument suggesting that hypertensive heart disease (HHD) may not represent a true pathological entity but rather a coding construct. While the author's intent to promote conceptual rigor is appreciated, the essay overlooks well-established pathophysiological mechanisms linking sustained hypertension to progressive cardiac structural and functional alterations.

The author acknowledges that hypertension induces left ventricular hypertrophy (LVH), yet paradoxically defines this remodeling as "paraphysiological". Such reasoning disregards the mechanical basis of hypertrophy as a response to chronically increased afterload. The relationship between wall stress ($\sigma = P \cdot r / 2h$) and compensatory myocyte growth is a hallmark of cardiac adaptation to pressure overload [2]. An increase in arterial impedance—due to reduced arterial compliance, elevated arterial pressure, or increased peripheral resistance—further raises ventricular wall stress and accelerates remodeling. However, this adaptive process is neither benign nor indefinitely stable: progressive interstitial fibrosis, impaired relaxation, and elevation of left ventricular end-diastolic pressure (pD_2) inevitably ensue [3, 4].

The conceptual distinction between adaptation and disease is gradual—as proposed in Pickering's "continuum" concept [5]—not dichotomous. LVH secondary to hypertension represents the initial step in a well-documented cascade leading to diastolic dysfunction, left atrial enlargement, and ultimately heart failure with preserved ejection fraction (HFpEF). Large epidemiological studies, including the Framingham cohort, have demonstrated that increased LV mass independently predicts cardiovascular mortality and arrhythmic events, irrespective of clinical symptoms [6, 7].

Elevation of LV end-diastolic pressure promotes left atrial dilatation, predisposing to electrical instability and atrial fibrillation [8, 9]. Such changes represent markers of chronic maladaptation strongly associated with adverse outcomes, rather than benign or quasi-physiological remodeling.

Diseases need not follow an invariable trajectory to be recognized as pathological entities. Atherosclerosis, for instance, exhibits wide interindividual variability yet remains a cornerstone of clinical cardiology. Likewise, hypertensive heart disease should be understood as a progressive structural and functional continuum rather than a binary condition [10]. The absence of symptoms does not imply the absence of pathology.

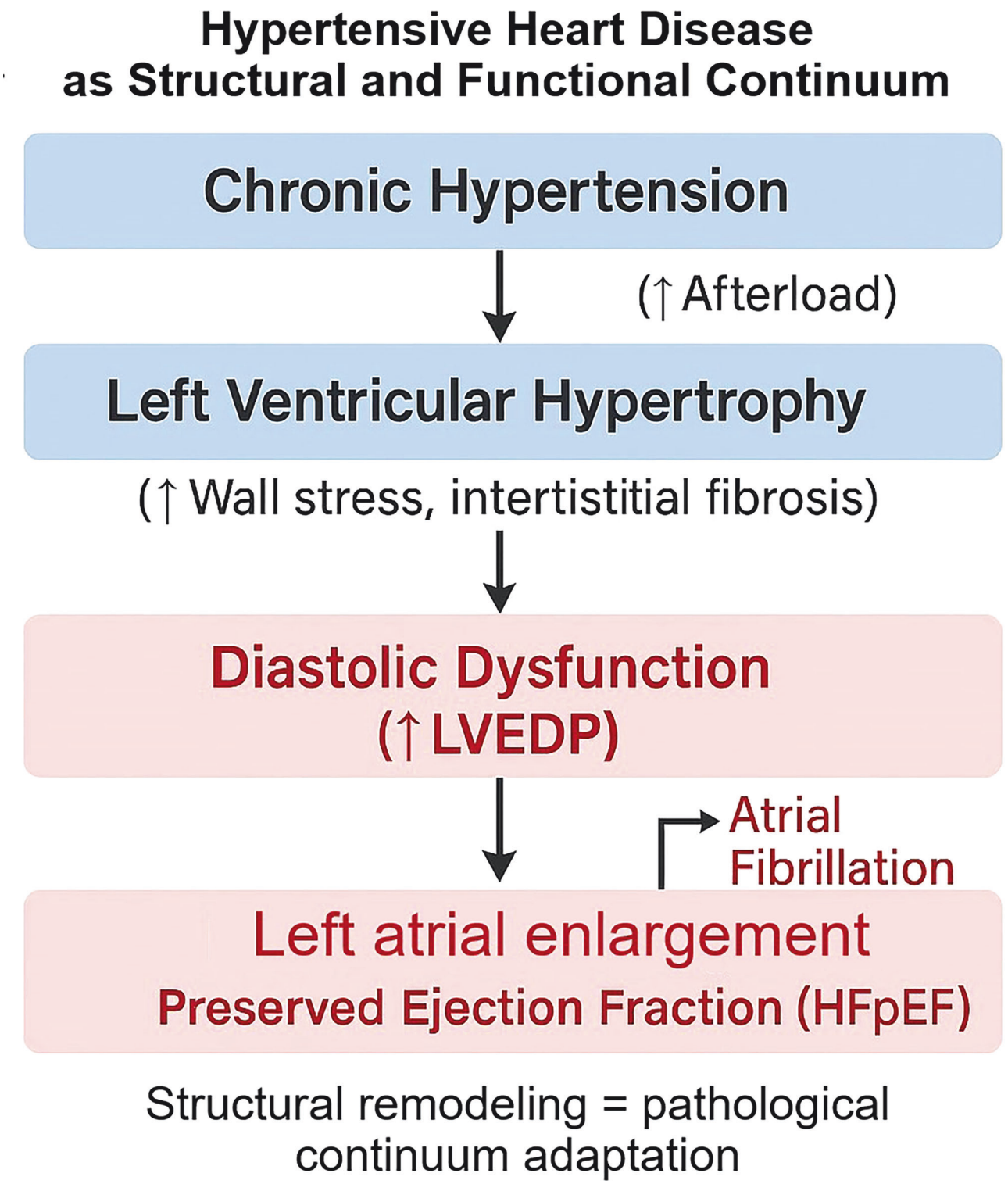
Left ventricular hypertrophy and left atrial enlargement are not benign findings. They are predictors of cardiovascular morbidity and mortality. Labeling them as "within normal limits for aging" risks trivializing essential diagnostic information, undermining preventive cardiology, and promoting underrecognition of subclinical disease [11].

The proposal to abandon the term "hypertensive heart disease" is therefore both conceptually and clinically debatable. Rather than eliminating the term, efforts should focus on refining diagnostic criteria to reflect the pathophysiological continuum of cardiac remodeling secondary to chronic pressure overload. "Hypertensive heart disease" remains a valid and clinically meaningful construct that captures the transition from adaptive hypertrophy to

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Graphical Opinion



functional impairment—a process deeply rooted in cardiovascular physiology.

Respectfully,

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Compliance with ethical standards

Conflict of interest The author declares no competing interests.

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